

30 Days Frontend Development (HTML + CSS + JavaScript + React JS)

Duration: 30 Days

Daily Class: 3 Hours

Total Hours: 90 Hours

Module 1 – HTML (Days 1–6)

Day 1 (3 Hours)

Topics

- Introduction to Web Development
- Frontend vs Backend
- HTML Basics
- VS Code Setup
- HTML Structure
- Live Server
- First Webpage

Duration Allocation

- Introduction to Web Development & Frontend Concepts – **45 Minutes**
 - HTML Fundamentals & Development Environment Setup – **45 Minutes**
 - Live Coding: HTML Structure & First Webpage – **60 Minutes**
 - Hands-on Practice, Discussion & Q&A – **30 Minutes**
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Day 2 (3 Hours)

Topics

- Headings
- Paragraphs
- Text Formatting
- Comments
- Lists
- Links
- Images

Mini Task

Duration Allocation

- HTML Content Elements (Theory) – **45 Minutes**

- Live Coding Demonstration – **60 Minutes**
 - Hands-on Practice & Mini Task – **75 Minutes**
-

Day 3 (3 Hours)

Topics

- Tables
- Forms
- Input Types
- Semantic HTML
- Audio
- Video
- iframe

Mini Project: Student Registration Form

Duration Allocation

- Theory & Concepts – **45 Minutes**
 - Live Coding Demonstration – **45 Minutes**
 - Mini Project Development – **90 Minutes**
-

Day 4 (3 Hours)

Topics

- HTML5 Semantic Tags
- Header
- Footer
- Section
- Article
- Aside
- Navigation

Mini Project

Duration Allocation

- Semantic HTML Concepts – **45 Minutes**
- Live Demonstration – **45 Minutes**
- Practical & Mini Project Development – **90 Minutes**

Day 5 (3 Hours)

Topics

- SEO Basics
- Meta Tags
- Favicon
- Best Practices
- Accessibility Basics

Practice

Duration Allocation

- Theory & Best Practices – **60 Minutes**
 - Live Demonstration – **45 Minutes**
 - Practical Exercises – **75 Minutes**
-

Day 6 (3 Hours)

HTML Project: Portfolio Website (HTML Only)

Duration Allocation

- Project Planning – **30 Minutes**
 - Project Development – **120 Minutes**
 - Project Review & Feedback – **30 Minutes**
-

Module 2 – CSS (Days 7–13)

Day 7 (3 Hours)

Topics

- CSS Introduction
- Inline
- Internal
- External
- Selectors

Duration Allocation

- CSS Fundamentals – **45 Minutes**
- Live Coding – **60 Minutes**

- Hands-on Practice – **75 Minutes**
-

Day 8 (3 Hours)

Topics

- Colors
- Backgrounds
- Fonts
- Text Styling
- Icons

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Practical Session – **75 Minutes**
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Day 9 (3 Hours)

Topics

- Box Model
- Margin
- Border
- Padding
- Width
- Height

Duration Allocation

- Theory – **45 Minutes**
 - Live Demonstration – **60 Minutes**
 - Hands-on Practice – **75 Minutes**
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Day 10 (3 Hours)

Topics

- Display
- Position
- Overflow

- Float
- Z-index

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Practical Session – **75 Minutes**
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Day 11 (3 Hours)

Topics

- Flexbox
- Alignment
- Responsive Layout

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Responsive Layout Practice – **75 Minutes**
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Day 12 (3 Hours)

Topics

- CSS Grid
- Media Queries
- Responsive Design

Duration Allocation

- Theory – **45 Minutes**
 - Live Demonstration – **60 Minutes**
 - Practical Exercises – **75 Minutes**
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Day 13 (3 Hours)

CSS Project: Responsive Portfolio Website

Duration Allocation

- Project Planning – **30 Minutes**
- Project Development – **120 Minutes**

- Project Review & Feedback – **30 Minutes**
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Module 3 – JavaScript (Days 14–20)

Day 14 (3 Hours)

Topics

- JavaScript Introduction
- Variables
- Data Types
- Operators

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Practical Exercises – **75 Minutes**
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Day 15 (3 Hours)

Topics

- Conditions
- Loops
- Functions
- Scope

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Hands-on Practice – **75 Minutes**
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Day 16 (3 Hours)

Topics

- Arrays
- Objects
- Array Methods
- String Methods

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Practical Exercises – **75 Minutes**
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Day 17 (3 Hours)

Topics

- DOM
- Selecting Elements
- Events
- Event Listeners

Duration Allocation

- Theory – **45 Minutes**
 - Live Demonstration – **60 Minutes**
 - Practical Exercises – **75 Minutes**
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Day 18 (3 Hours)

Topics

- Form Validation
- Local Storage
- JSON
- Session Storage

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Practical Exercises – **75 Minutes**
-

Day 19 (3 Hours)

Topics

- ES6 Features
- Arrow Functions
- Destructuring
- Spread

- Rest
- Modules
- Template Literals

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Hands-on Practice – **75 Minutes**
-

Day 20 (3 Hours)

JavaScript Project: Todo Application

Duration Allocation

- Project Planning – **30 Minutes**
 - Project Development – **120 Minutes**
 - Testing & Review – **30 Minutes**
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Module 4 – React JS (Days 21–30)

Day 21 (3 Hours)

Topics

- React Introduction
- What is React?
- Why React?
- Virtual DOM
- SPA
- React Architecture
- Install Node.js
- npm
- Vite
- Create React App

Lab: Hello React

Duration Allocation

- React Fundamentals – **45 Minutes**
- Environment Setup & Live Coding – **60 Minutes**

- Lab Exercise – **75 Minutes**
-

Day 22 (3 Hours)

Topics

- JSX
- Functional Components
- Component Structure
- Export & Import
- Folder Structure

Project: Profile Card

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
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Day 23 (3 Hours)

Topics

- Props
- useState
- Dynamic Rendering
- Event Handling

Project: Counter App

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
-

Day 24 (3 Hours)

Topics

- useEffect
- API Calls
- Fetch

- Axios

Project: User List Application

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
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Day 25 (3 Hours)

Topics

- Controlled Components
- Validation
- Multiple Inputs

Project: Registration Form

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
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Day 26 (3 Hours)

Topics

- React Router
- BrowserRouter
- Routes
- Link
- useNavigate
- useParams

Project: Multi-Page Website

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
-

Day 27 (3 Hours)

Topics

- GET
- POST
- PUT
- DELETE
- Axios
- JSON Server

Project: Student Management System

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
-

Day 28 (3 Hours)

Topics

- createContext
- useContext
- Theme Switcher
- Global State

Project: Dark & Light Theme

Duration Allocation

- Theory – **45 Minutes**
 - Live Coding – **60 Minutes**
 - Project Development – **75 Minutes**
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Day 29 (3 Hours)

Topics

- Custom Hooks
- useRef
- useMemo
- React.memo

- Lazy Loading
- Folder Structure
- Deployment (Vercel/Netlify)

Duration Allocation

- Theory – **45 Minutes**
 - Live Demonstration – **60 Minutes**
 - Practical Implementation – **75 Minutes**
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Day 30 (3 Hours)

Final Capstone Project: Student Management Portal

Features

- Login Page
- Dashboard
- Navigation
- CRUD Operations
- Search
- Filter
- Responsive Design
- API Integration
- Form Validation
- Local Storage
- Theme Switcher
- Protected Routes (Basic)
- Deployment

Duration Allocation

- Project Planning & Architecture – **30 Minutes**
- Capstone Project Development – **120 Minutes**
- Project Presentation, Evaluation & Feedback – **30 Minutes**